Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21307

F-407

B.C.A. (Second Semester)

Examination, 2024

(New Course)

Paper No. BCA-N-202

DISCRETE MATHEMATICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Differentiate between conjunction and disjunction. Provide an example for each.

   (b) Discuss the duality of conditional and unconditional statements.

2. (a) Prove that if a graph (connected or diconnented) has exactly two vertices of odd degree, then there exist a path joining these two vertices.

   (b) Prove that every connected graph with n vertices and n−1 edges is a tree.

3. (a) Prove that a Boolean algebra is a self dual.

   (b) Show that the union of two equivalence relation need not be an equivalence relation.

4. (a) Show that:

   (a) $\sim(p \vee q) \vee (\sim p \wedge q) \vee p$ is a tautology

   (b) $((p \vee q) \wedge \sim q) = p$ is a tautology

   (c) $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradication.

   (b) Minimize the four variable logic function using K-Map

   f(A, B, C, D) = A B C'D + A'BCD + A'B'C' + A'B'D' + AC' + AB'C + B'

5. (a) Simplify the expression (x+y) (x+z) using the laws of boolean algebra.

   (b) Using Boolean identities, reduce the given Boolean expression:

   F(X, Y, Z)=X'Y+ YZ' + YZ + XY'Z'

6. (a) Let A= {0, 1, 2, 3, 4} and define a relation R on A as follows:

R={(0, 0), (0, 4), (1, 1) (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)}.

R is an equivalence relation on A. Find the distinct equivalence classes of R.

(b) Prove that the following argument is valid:

"all dogs are carnivorous."

"some animals are dogs."

Therefore" some animals are carnivorous".

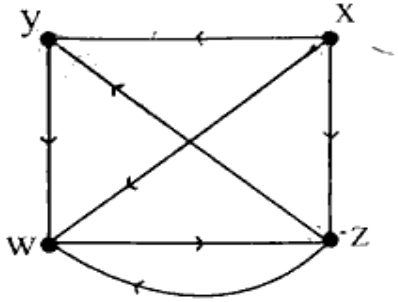
7. (a) Construct the binary tree whose inorder and preorder traversal is given below.

Also, find the postorder traversal of the tree.

Inorder : d, g, b, e, i, h, j, a, c, f

Preorder : a, b, d, g, e, h, i, j, c, f

(b) Consider the directed graphy G in figure:



(a) Find indegree and out degree of each vertex of G.

(b) Find all simple paths from y to z

(c) Find all cycles in G.

8. (a) Use truth table to prove the following argument

$P \rightarrow \sim q$

$r \rightarrow q$

$\frac{q}{\therefore \sim r}$

(b) Let g: A→B and f:B→C be two functions. Show that if g and f are both injectives, then $f \circ g : A \rightarrow C$ is injective.